Combining Temporal, Environmental, Demographic, and Geospatial Information to Forecast Influenza by Neighborhood: A Graph Neural Network in 3 US Cities

Abstract

Health departments act by neighborhood, but most influenza forecasts cover states or regions. We asked whether a graph neural network combining each neighborhood's recent trajectory, weather, demographic measures, and a map of neighboring areas improves neighborhood forecasts, and which part accounts for any gain. We forecast emergency department visits for influenza-like illness 1, 2, and 4 weeks ahead in Boston, Massachusetts (14 neighborhoods, 2022–2025); Columbus, Ohio (17 areas, 2024–2025); and Chicago, Illinois (58 ZIP codes, 2022–2026), on seasons not used in design, with equal tuning, scored by the weighted interval score. In Boston, the model had 18% to 28% less error than persistence (next week equals this week); a tuned recurrent neural network matched it 1 week ahead. Removing the map raised error by 4% to 11% in Boston and 4% to 7% in Chicago, clearly at 1 week in every season; removing weather or demographics changed error by under 4%. Maps from public boundary files matched a hand-drawn one, and a Boston-trained model forecast Columbus as well as a local model. Of the 3 components, only the map clearly lowered error.

Keywords: forecasting; graph neural networks; influenza, human; machine learning; public health surveillance; small-area analysis

Abbreviations: ARIMA, autoregressive integrated moving average; ED, emergency department; ILI, influenza-like illness; LSTM, long short-term memory network; WIS, weighted interval score.

Introduction

In Boston, the average weekly rate of emergency department (ED) visits for influenza-like illness (ILI) was more than 7 times higher in Dorchester (67 per 100,000 residents) than in Fenway (9 per 100,000) between June 2022 and May 2025.[1] Neighborhoods within a single city differ in age structure, income, and housing, and these differences have long been tied to who becomes ill with influenza.[2–5] Yet neighborhoods also sit side by side in a densely populated, highly connected city, where residents commute, work, and shop across neighborhoods[6–8] and where influenza spreads between nearby places.[9,10]

Health departments act on a community level. Mobile vaccination clinics, school outreach, and messaging to clinicians are directed to particular neighborhoods and schools,[11–13] often before the data that would guide these decisions become available.[14] Forecasts can help inform these choices,[15] yet most influenza forecasting targets nations, regions, and states.[16–18] Forecasting within a city is harder because each area has fewer cases and noisier weekly counts,[19,20] and an earlier study in New York City found that linking areas improved borough forecasts but not neighborhood forecasts.[21]

Neighborhood influenza activity is plausibly shaped by several kinds of information at once: its own recent trajectory and seasonal timing,[22,23] weather conditions that affect transmission,[24–26] the demographic makeup of the neighborhood,[5,27,28] and activity in neighboring areas.[10,21] Forecasting models have rarely combined them: in a review of 35 validated influenza forecasting studies, models drew mainly on surveillance series, 6 used weather, and none reported demographic or neighboring-area inputs.[29] Graph neural networks, which pass information along a map of connections, offer a way to combine all of them in a single model.[30,31] Graph models have reported gains for influenza across regions[32–35] and for COVID-19 across German districts,[36] but the evidence is mixed. In forecasting hubs, many models failed to beat simple baselines in some seasons;[18,37] autoregressive models remain competitive 1 to 4 weeks ahead;[23] and in a study of dengue in Sri Lanka, a recurrent neural network fitted to each district outperformed every graph variant.[38] Reviews also caution that

the added value of artificial intelligence for epidemics is hard to judge[39] and that retrospective tests can overstate real-world accuracy.[40]

It is therefore unclear whether combining temporal, environmental, demographic, and geospatial information in a graph neural network improves neighborhood-level forecasts in a city like Boston, and where any gain comes from. We addressed this question by forecasting neighborhood ED visit rates for ILI 1, 2, and 4 weeks ahead in Boston's 14 neighborhoods. We asked 1) whether the combined model outperforms simple and machine learning comparison models; 2) how much each component (environmental, demographic, and geospatial) adds to its forecasts; and 3) whether the map can be built from public boundary files and whether a model trained in one city can forecast another. To limit the usual ways a new model can look better than it is, we tested on seasons never used in design,[41] gave every model the same tuning budget, and scored full forecast distributions.[42,43] We repeated the analysis in Columbus, Ohio, and in Chicago, Illinois, a city never used while designing the model.

This study makes 4 contributions. First, it evaluates neighborhood-level influenza forecasts in a way designed to guard against overstated gains. Second, it separates what the environmental features, the demographic features, and the neighborhood map each add to a graph neural network, on seasons not used to design it. Third, it tests whether the map can be built automatically from public boundary files and whether a model trained in one city transfers to another, in Boston, Columbus, and Chicago. Fourth, to support reproducibility, we release the code and frozen settings in a public repository,[44] along with an open-source Python package that runs the forecasts and the evaluation on a health department's own data.[45]

Methods

Setting and data

Table 1 summarizes the 3 cities. The Boston Public Health Commission publishes the weekly rate of ED visits for ILI per 100,000 residents for 14 neighborhoods;[1] ED visits track laboratory-confirmed influenza closely, including by age group.[46] Suppressed weeks with small counts were treated as missing, not zero. Columbus Public Health provided visit-level ED records, which we aggregated to 17 planning areas and divided by American Community Survey population.[47] The Chicago Department of Public Health publishes a weekly ILI activity level from 1 to 10 for each ZIP code,[48] so Chicago errors are in levels (Appendix S1).

Table 1: Surveillance Data and Evaluation Seasons for Neighborhood-Level Influenza Forecasting in Boston, Massachusetts; Columbus, Ohio; and Chicago, Illinois, 2017–2026

Characteristic	Boston	Columbus	Chicago
Areas forecast	14 neighborhoods	17 planning areas	58 ZIP codes
Surrounding areas, not scored	7	5	1
Measure	ED visits for ILI per 100,000	ED visits for ILI per 100,000	ILI activity level, 1–10[a]
Source	Boston Public Health Commission	Columbus Public Health, visit-level records	Chicago Department of Public Health
Population denominator	Published rate	American Community Survey	None (index)
Data begin	December 2017	January 2022[b]	September 2019
Last week used	May 2026	May 2026	May 2026
Peak months	October–March	October–March	October–March
Test year	June–May	June–May	June–May
Unseen test seasons	2022–2023, 2023–2024, 2024–2025	2024–2025	2022–2023 to 2025–2026
Development season[c]	2025–2026	2025–2026	None
Neighborhood map	Hand drawn; also built from boundary files	Built from boundary files	Built from boundary files
Demographic measures	8	8	8

Abbreviations: ED, emergency department; ILI, influenza-like illness.

[a] A ZIP code's share of ED visits for ILI against its own summer average, in steps of about 1 standard deviation and capped at 10.

[b] Columbus models used weeks from July 2022 onward.

[c] Used to design the model and not reported as evidence.

Boston and Chicago data are public weekly aggregates; Columbus records were deidentified and aggregated to weekly area counts before analysis.

Four kinds of information

The model received 4 kinds of information for each area and week (Figure 1C). *Temporal*: the area's rate over the past 16 weeks, a flag for suppressed weeks, and its usual rate for the time of year (in the baseline below). *Environmental*: 6 weekly weather variables, including temperature and humidity, from the Open-Meteo archive,[49] because humidity and temperature shape influenza transmission.[24–26] *Demographic*: 8 fixed measures, including poverty, age structure, and housing, because age and socioeconomic position shape who becomes ill[4,5,27] (Appendix S1). *Geospatial*: a map of how areas connect, described next.

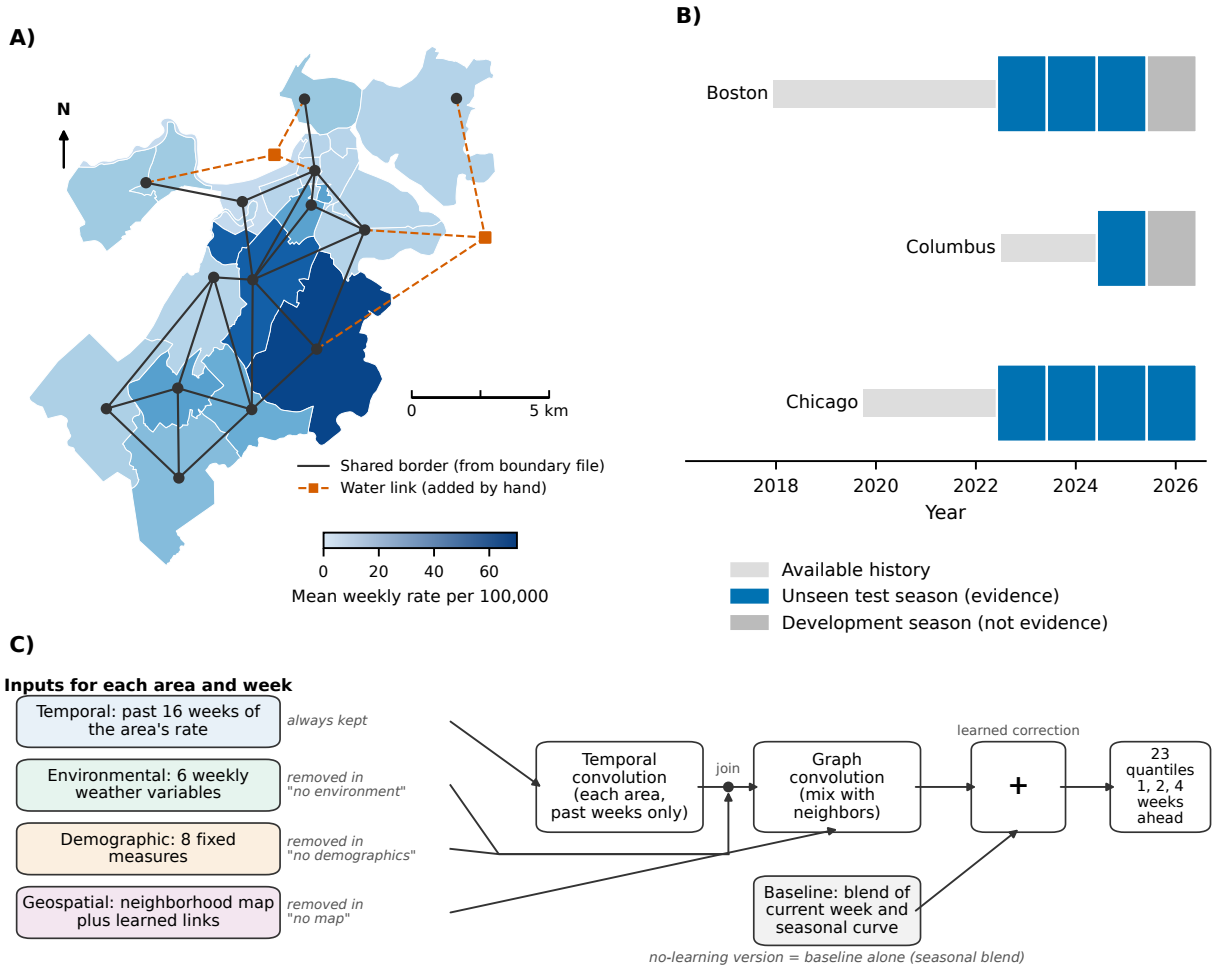


Figure 1. Study design and model for neighborhood-level influenza forecasting in Boston, Massachusetts; Columbus, Ohio; and Chicago, Illinois, 2017–2026. A) Boston's 14 neighborhoods, shaded by mean weekly rate of emergency department visits for influenza-like illness per 100,000 residents, June 2022–May 2025. Solid lines link neighborhoods that share a border in the official land boundary file; dashed lines are links across the Charles River and Boston Harbor, which were added by hand. Other surrounding areas are not shown. The evaluated model used the hand-drawn map, which differs from this one in 5 of its 23 borders. B) Data used in each city and the test seasons. Each model was trained only on data before a test season and scored on its target weeks. The development season, used to design the model, is not reported as evidence; Chicago, added after the design was fixed, has none. C) The graph neural network. Each area's past 16 weeks of rates enter a temporal convolution that looks only at past weeks; the current week's weather and the demographic measures are joined to its summary; and the neighborhood map enters at the graph convolution, which mixes each area's summary with its neighbors'. The model's learned correction is added to a baseline that blends the current week with the seasonal curve, and the output is 23 quantiles of the forecast distribution. Italic notes give the version of the model in which each input is removed (Figure 5).

Alt text: Top left, a map of Boston's 14 neighborhoods shaded from light to dark blue, darkest in Dorchester and Roxbury, with dots at each neighborhood connected by lines to bordering neighborhoods and dashed lines to 2 water points. Top right, horizontal bars for Boston, Columbus, and Chicago showing years of available data, blue blocks for unseen test seasons (3 for Boston, 1 for Columbus, 4 for Chicago), and gray blocks for the Boston and Columbus development seasons. Bottom, a flow diagram: 4 input boxes (temporal, environmental, demographic, geospatial) on the left; the temporal box points to a temporal convolution box, the environmental and demographic boxes join its output, and the geospatial box points to a graph convolution box; the graph convolution and a baseline box both point to a plus sign, which points to an output box of 23 quantiles 1, 2, and 4 weeks ahead.

The neighborhood map

Each city is a map in which every area is a point and 2 areas are linked when they share a border (Figure 1A). So that areas at the edge of a city do not look as if they had no neighbors on one side, the map also includes surrounding areas that are never forecast or scored (7 in Boston, 5 in Columbus, and 1 in Chicago; Appendix S1).

Boston's map was first drawn by hand. To test whether a new city could skip that step, we also built it from public boundary files,[50–52] treating 2 areas as neighbors when their boundaries share an edge rather than only a corner. Building neighbor lists from census boundary files is standard in spatial disease mapping, as is adding links across water by hand,[53] as we did for the Charles River and Boston Harbor. Besides shared borders, which also reach 2 and 3 borders away at decreasing weight, the default map links neighborhoods whose past influenza curves were highly correlated (before the test season only) and neighborhoods with similar demographic measures (weights in Appendix S1).

Forecasting model

The graph neural network forecasts each area in 3 steps (Figure 1C). First, a temporal convolution summarizes the area's rate and suppression flag over the past 16 weeks, and the current week's weather and the demographic measures are joined to that summary. Second, a graph convolution[30] mixes each area's summary with its neighbors'. With $\mathbf{H}$ the summaries of all areas stacked as rows,

$$\mathbf{H}' = \mathrm{LayerNorm}\Big(\mathbf{H} + \mathrm{ReLU}\big(g_1\,\tilde{\mathbf{A}}\mathbf{H}\mathbf{W}_1 + g_2\,\mathbf{B}\mathbf{H}\mathbf{W}_2\big)\Big),$$

where $\tilde{\mathbf{A}}$ is the normalized table of map links (shared borders, correlated curves, and similar demographics, combined into one set of weights), $\mathbf{B}$ is a table of links the model learns for itself, $\mathbf{W}_1$ and $\mathbf{W}_2$ are learned, and the learned weights g_1 and g_2 sum to 1. This step is applied twice. Third, the model adds a learned correction $r_{i,h}$ to a baseline that blends 2

simple forecasts: the current week's rate $y_{i,t}$ and the usual rate for the target week $c_i(t+h)$, estimated from training seasons with a smooth seasonal curve,

$$\hat{y}_{i,t+h} = \pi_h \, y_{i,t} + (1 - \pi_h) \, c_i(t+h) + r_{i,h},$$

with $0 < \pi_h < 1$ learned for each horizon h (weeks ahead; Appendix S1, Figure S1). Each area's series was scaled by its training-period mean and standard deviation. A separate model was trained for each horizon, and each forecast averages 10 training runs.

Each model produced a full forecast distribution, reported as the 23 quantiles (percentiles) FluSight uses,[18] with ranges from one procedure, shared by all models except the FluSight baseline, calibrated on the weeks just before each test season.

Comparison models

Four comparison models use only each area's own history: persistence (next week equals this week); the FluSight baseline, which carries the latest value forward with ranges from past changes;[18,37] the seasonal blend, which is the graph neural network's baseline with no learned correction ($r_{i,h} = 0$); and a seasonal autoregressive integrated moving average (ARIMA) model for each area. Three are machine learning models: gradient-boosted regression trees on the graph neural network's inputs except the map; a long short-term memory network (LSTM), a recurrent neural network here reading all areas' series together but with no map;[54] and Cola-GNN, a published influenza graph neural network that learns its own connections between locations,[32] which we reimplemented. Appendix S1 describes these and 4 further models.

What each component adds

To measure what each kind of information contributes, we retrained an otherwise identical model with one removed: 1) no environment, without the 6 weather variables; 2) no de-

mographics, without the 8 demographic measures as inputs (links between demographically similar areas belong to the map and were kept); 3) no map, with no links between areas, mapped or learned, and no surrounding areas, so each area sees only its own inputs; and 4) none of the three, which sees only each area's own past rates. Each kept the same baseline. For component c, the change in error when it is removed is

$$\Delta_c = 100 \times \frac{\overline{\text{WIS}}_{-c} - \overline{\text{WIS}}_{\text{full}}}{\overline{\text{WIS}}_{\text{full}}},$$

where $\overline{\text{WIS}}$ is the mean weighted interval score defined below; $\Delta_c > 0$ means the component lowered error. If the components are what separate the graph neural network from history-only methods, the version with none of them should forecast about as well as those methods. Our criterion, set before the results were seen, was relative WIS within 5% of the seasonal blend's, and no clear difference from it (defined below) in most season and horizon comparisons.

Evaluation design

In rolling-origin evaluation,[41] every model was trained only on data before each test season, tuned on the weeks just before it, and scored on its 52 target weeks (53 in Chicago 2025–2026).

The 2025–2026 season in Boston and Columbus was used to design the model (the development season; Appendix S1) and is not reported as evidence. The other seasons were never used in design: 2022–2023, 2023–2024, and 2024–2025 in Boston and 2024–2025 in Columbus. Chicago was added after the design was fixed, so all 4 of its seasons, 2022–2023 through 2025–2026, are unseen (Figure 1B). Influenza nearly disappeared during the first year of the COVID-19 pandemic,[55] so Columbus models used weeks from July 2022 onward, leaving 1 unseen season; Boston and Chicago models used their full histories.

Every machine learning model, including each version with a component removed, got

the same tuning budget in every city, test season, and horizon: 12 combinations of settings (its default plus 11 drawn at random), each trained once, with the best on the validation weeks retrained 10 times (Appendix S1).

Scoring

Our main measure answers the question: How far off was the forecast, counting both its best guess and how honest its range was? This is the weighted interval score (WIS), the scoring rule used by FluSight and the COVID-19 Forecast Hubs.[18,43] Each forecast gives a median m and $K = 11$ central intervals $[l_k, u_k]$ (98% to 10%); for observed value y,

$$\mathrm{WIS} = \frac{1}{K + \frac{1}{2}}\left(\tfrac{1}{2}|y - m| + \sum_{k=1}^{K}\left[\tfrac{\alpha_k}{2}(u_k - l_k) + (l_k - y)^+ + (y - u_k)^+\right]\right),$$

where α_k is 1 minus the interval's coverage and $(x)^+ = \max(x, 0)$: interval widths plus penalties for observations outside them (Appendix S1). It is in the units of the forecast (ED visits per 100,000, or levels in Chicago), lower is better, and it rewards ranges that are narrow but still contain what happened. To compare across cities and horizons, relative WIS is $\overline{\mathrm{WIS}}_{\mathrm{model}}/\overline{\mathrm{WIS}}_{\mathrm{persistence}}$, where $\overline{\mathrm{WIS}}$ averages over areas and target weeks within each season and then over a city's unseen seasons; it is below 1 when the model beats persistence. Mean absolute error, root mean squared error, and the correlation between forecast and observed rates score only the best guess, so they are secondary. We checked calibration by asking how often the observed value fell inside the 50% and 95% forecast intervals.[42]

To compare 2 models within a season, we asked whether the gap between them could be explained by which weeks the season happened to contain. We averaged WIS over areas for each week, resampled blocks of 4 consecutive weeks for a 95% confidence interval (2,000 resamples),[56] and used the Diebold–Mariano test,[57] using a small-sample correction[58] and a variance estimate that allows for the overlap between consecutive multiweek forecasts.[59] Two-sided P values were Holm-adjusted[60] within each season and horizon, separately for the

comparison models, the 4 versions with a component removed, and the history-only check. We call a difference *clear* when the adjusted P value was below .05.

We also scored alerts for moderate and high weeks, with neighborhood thresholds set by the Moving Epidemic Method[61] on earlier seasons as in the Centers for Disease Control and Prevention's severity classification,[62] using the Brier skill score[63] for the forecast's probability of crossing a threshold and hit rate minus false-alarm rate (Peirce skill score) for its best guess (Appendix S1).

Portability

With model settings fixed in advance, we reran the Boston evaluation with 2 automatically built maps, for the graph neural network and for a borders-only version whose map is its only spatial information. We also trained a version with no neighborhood-specific learned links, which can run in any city, on one city's weeks before 2024–2025 and applied it to another city's 2024–2025 season as is and after brief further training on local data (fine-tuning): from Boston to Columbus and Chicago, and from each to Boston. We compared it with the same model trained from scratch there, and also trained it on 2 or 3 cities at once (Appendix S1).

Analyses used Python 3.12, PyTorch 2.14, PyTorch Geometric 2.8, statsmodels 0.14, scikit-learn 1.9, and XGBoost 3.4; the code that reproduces every run is released.[44]

We used generative artificial intelligence (Claude Opus 4.5, 4.6, 4.7, 4.8, 5, and 5.5; Anthropic)[64–69] through Claude Code to help write and debug code, run the analyses we specified, find and check references, edit text, and cross-check numbers against output files. The authors designed the study; chose the data, comparisons, and evaluation criteria; drafted the text; reviewed all code, outputs, and text; and take full responsibility for the content.

Results

How well the combined model forecasts

In Boston, the graph neural network had the lowest mean WIS of the comparison models at every horizon (Table 2): 5.13 per 100,000 at 1 week, 6.47 at 2 weeks, and 8.22 at 4 weeks, against 6.24, 7.94, and 11.44 for persistence, or 18% to 28% less error (Figure 2A). This advantage was clear in all 3 seasons at 1 week and in 2023–2024 at 2 and 4 weeks (Figure 3A). Point measures largely agreed (Appendix S1, Table S10).

Table 2: Forecast Error by Model and Horizon for 14 Neighborhoods, Boston, Massachusetts, 3 Unseen Seasons, June 2022–May 2025

	Mean WIS[a]			Relative WIS[b]			Mean Absolute Error[a]		
Model	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk
Graph neural network	5.13	6.47	8.22	0.82	0.81	0.72	7.85	9.63	11.93
Same model, map removed	5.69	6.72	8.68	0.91	0.85	0.76	8.76	10.10	12.57
LSTM	5.18	6.82	9.16	0.83	0.86	0.80	7.96	10.19	13.02
Cola-GNN	5.35	7.61	8.77	0.86	0.96	0.77	8.19	11.04	12.27
Gradient-boosted trees	6.08	7.32	9.05	0.97	0.92	0.79	9.21	10.82	12.99
Seasonal ARIMA	6.52	7.66	9.15	1.04	0.97	0.80	9.69	11.17	13.09
Seasonal blend, no learning	5.95	7.29	9.57	0.95	0.92	0.84	9.06	10.88	13.69
FluSight baseline	6.39	8.16	11.45	1.02	1.03	1.00	9.50	11.89	16.23
Persistence	6.24	7.94	11.44	1.00	1.00	1.00	9.48	11.89	16.22

Abbreviations: ARIMA, autoregressive integrated moving average; ED, emergency department; LSTM, long short-term memory network; WIS, weighted interval score; wk, weeks ahead.

[a] ED visits for influenza-like illness per 100,000 residents; mean of the 3 season means; lower is better. Learned models were tuned with an equal budget and averaged over 10 training runs.

[b] Mean WIS divided by that of persistence; values below 1 indicate less error than persistence.

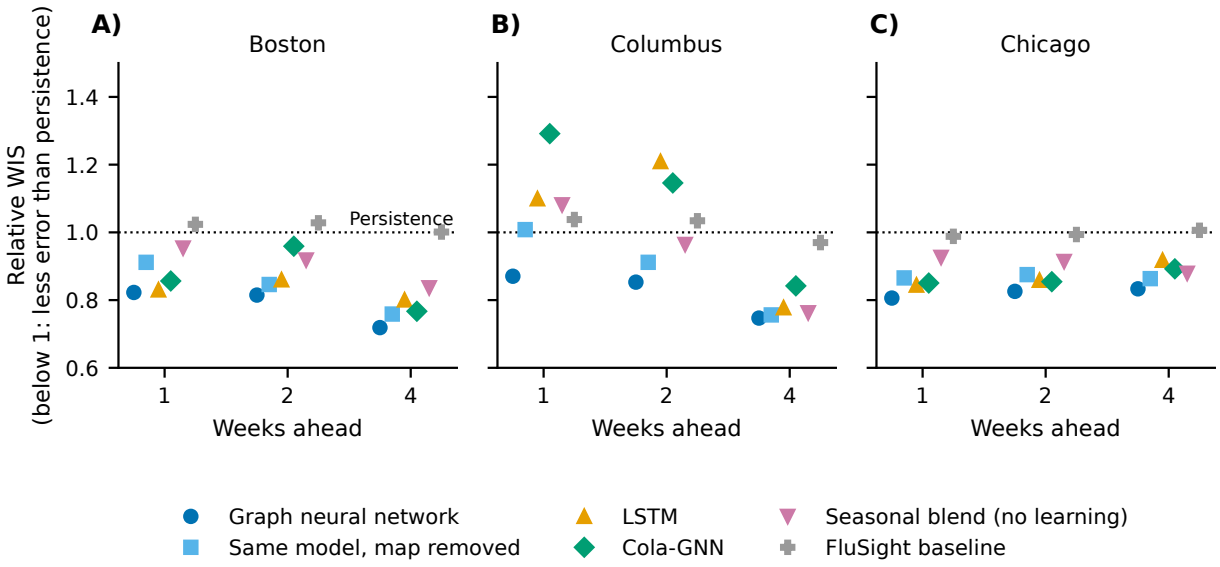


Figure 2. Forecast error relative to persistence by city, model, and horizon, unseen test seasons: A) Boston, Massachusetts, 2022–2025; B) Columbus, Ohio, 2024–2025; C) Chicago, Illinois, 2022–2026. Models shown are the graph neural network, the same model with the map removed, the tuned long short-term memory network (LSTM), tuned Cola-GNN, the seasonal blend (no learning), and the FluSight baseline. Relative weighted interval score (WIS) is each model's mean WIS divided by that of persistence (next week equals this week); values below the dotted line at 1 indicate less error than persistence. Learned models were tuned with an equal budget.

Alt text: Three dot plots, one per city, of relative error at 1, 2, and 4 weeks ahead for 6 models. In Boston and Columbus the graph neural network is lowest at every horizon, between 0.81 and 0.87 at 1 and 2 weeks and near 0.72 to 0.75 at 4 weeks. In Chicago it is lowest at every horizon, near 0.81 to 0.83, and the FluSight baseline sits near 1.

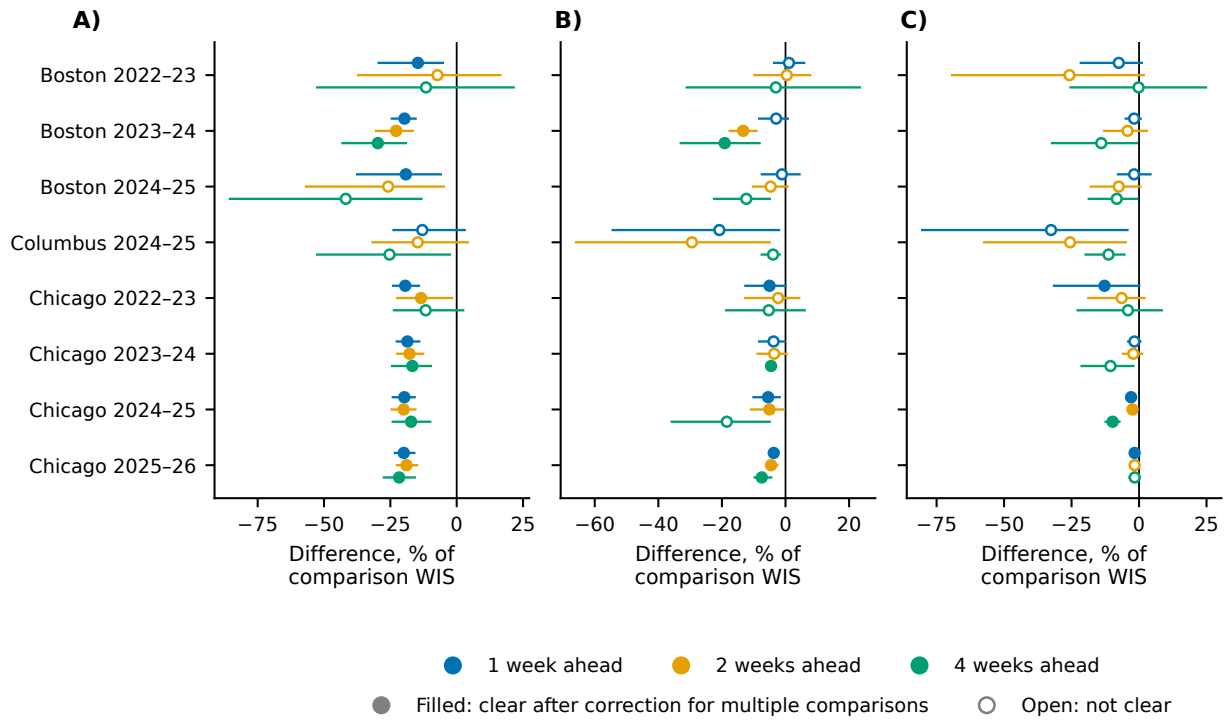


Figure 3. Week-level differences in forecast error between the graph neural network and 3 comparison models, by city, season, and horizon, Boston, Massachusetts (2022–2025); Columbus, Ohio (2024–2025); and Chicago, Illinois (2022–2026). A) Persistence. B) Tuned long short-term memory network (LSTM). C) Tuned Cola-GNN. Points show the difference in mean weighted interval score (WIS) (graph neural network minus comparison) as a percentage of the comparison model's mean WIS, so that Chicago's activity levels and the other cities' rates share one scale, and bars show 95% confidence intervals from a block bootstrap over weeks. Negative values favor the graph neural network. Filled points are differences with a Holm-adjusted 2-sided Diebold–Mariano P value below .05.

Alt text: Three forest plots with rows for Boston 2022–2023, 2023–2024, and 2024–2025, Columbus 2024–2025, and Chicago 2022–2023 to 2025–2026, and a point with an error bar for each of 3 horizons. Against persistence, nearly every point lies left of zero, filled for all Boston seasons at 1 week, for Boston 2023–2024 at every horizon, and for 11 of 12 Chicago points; Columbus points are open. Against the LSTM, Boston 1-week points sit on zero,

and Chicago points are left of zero and partly filled. Against Cola-GNN, most points lie left of zero; Boston and Columbus points are open, and the filled points are all in Chicago.

The margin over tuned machine learning models was smaller, and tuning helped them more than the graph neural network (across the 3 cities, validation error fell 7% for the LSTM and 2% for the graph neural network; Appendix S1, Figure S2). The tuned LSTM, which reads all 14 neighborhood series at once without a map, matched the graph neural network at 1 week in all 3 seasons (mean WIS 5.18 vs 5.13; Figure 3B). At 2 and 4 weeks the graph neural network had lower error (6.47 vs 6.82 and 8.22 vs 9.16), clearly only in 2023–2024. It also had lower average error than tuned Cola-GNN at every horizon, but no difference was clear (Figure 3C).

In Chicago, the graph neural network had the lowest mean WIS of the comparison models at every horizon (relative WIS 0.81, 0.83, and 0.83; Figure 2C), clearly beat persistence in 11 of 12 season-horizon comparisons, and, unlike in Boston, clearly beat the tuned LSTM and Cola-GNN at 1 week in 3 of 4 seasons (Figure 3). In Columbus, it had the lowest mean WIS of the models in Figure 2B (relative WIS 0.87, 0.85, and 0.75), but in this single season its lead over persistence, the LSTM, and Cola-GNN was not clear at any horizon, and at 4 weeks repeating the same week of the previous year had lower WIS (5.75 vs 6.31; Appendix S1, Tables S4–S9).

Forecast ranges were well calibrated: 93% to 94% of observed values fell inside the graph neural network's 95% intervals and 52% to 55% inside its 50% intervals (Figure 4A; Appendix S1, Figure S3). In Dorchester, the median forecast rose with the 2024–2025 epidemic earlier than persistence but fell short of the peak (Figure 4B–D; Appendix S1, Figure S4). The forecasts' probabilities warned of moderate weeks better than persistence at 1 and 2 weeks (Brier skill score 0.61 vs 0.51 and 0.41 vs 0.31), but because forecasts fell short of the highest peaks, yes/no alerts from the best guess caught fewer high weeks at 1 week (Peirce skill score 0.40 vs 0.59) and came no earlier (Appendix S1, Tables S13 and S14, Figures S5 and S6).

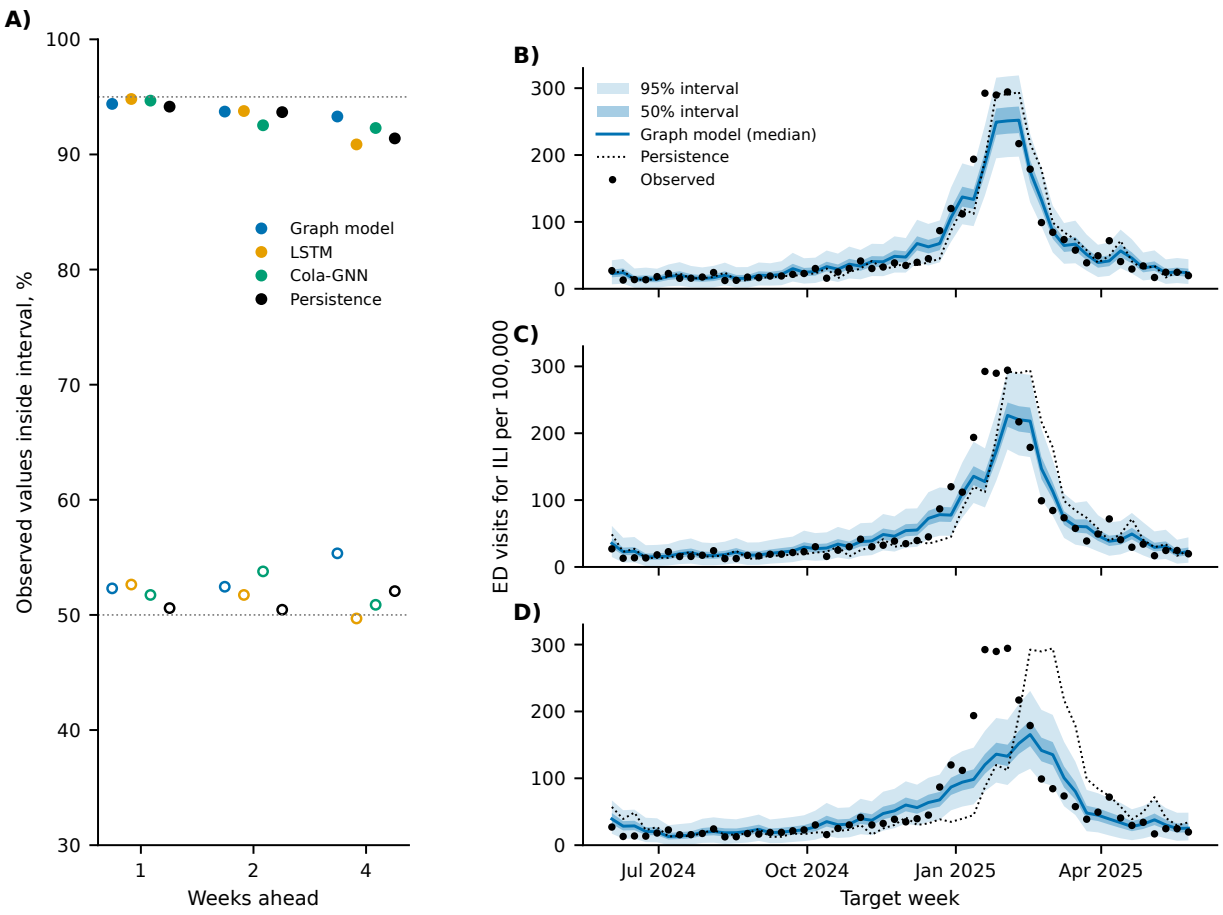


Figure 4. Example forecasts and calibration, Boston, Massachusetts, 2022–2025. A) Percentage of observed values inside the 50% (open circles) and 95% (filled circles) forecast intervals for the graph neural network, tuned long short-term memory network (LSTM), tuned Cola-GNN, and persistence by horizon, 14 neighborhoods, June 2022–May 2025. Dotted lines mark the nominal 95% and 50% coverage. B–D) Forecasts for Dorchester, the neighborhood with the highest rates, June 2024–May 2025, made 1 (B), 2 (C), and 4 (D) weeks ahead, showing the graph neural network's median forecast, 50% and 95% forecast intervals, persistence, and observed emergency department (ED) visit rates for influenza-like illness (ILI) per 100,000 residents. In Dorchester that season, relative weighted interval score (WIS; the graph neural network's mean WIS divided by that of persistence) was 0.85, 0.69, and 0.51 at 1, 2, and 4 weeks, against 0.82, 0.81, and 0.72 for all 14 neighborhoods over 3 seasons (Table 2), so panel D is a better-than-typical case.

Alt text: Left, coverage dots for 4 models cluster at 91% to 95% for the 95% intervals and 50% to 55% for the 50% intervals. Right, 3 stacked weekly time series for Dorchester from mid-2024 to mid-2025, with observed dots rising from about 13 to nearly 300 per 100,000 in early February 2025 and falling by April. At 1 week the model's median line follows the persistence line closely until the peak, where the median reaches about 250 and persistence about 290. At 2 weeks the median peaks near 225 and persistence trails 2 weeks behind. At 4 weeks the median peaks near 165, well below the observed peak, while persistence trails a month behind and stays near 290 into March, after the observed rates have fallen.

What each component adds

Which kinds of information account for the graph neural network's lead? Removing the map raised error the most (Table 3; Figure 5C; Appendix S1, Table S15). In Boston, error rose by 10.8% at 1 week, 3.8% at 2 weeks, and 5.6% at 4 weeks, and the increase was clear in all 3 seasons at 1 week and in 2 of 3 at 2 and 4 weeks. Averaged over the 3 seasons, the map lowered WIS in every one of the 14 neighborhoods at every horizon, by 6% to 15% at 1 week and by 2% to 8% at 2 and 4 weeks (Figure 6). In Chicago, removing the map raised error by 7.4%, 6.0%, and 3.6%, clearly at 1 week in all 4 seasons and at 2 weeks in 3, but not clearly at 4 weeks. In Columbus's single season, error rose by 15.8% at 1 week, not clearly.

Table 3: Change in Forecast Error When One Component Is Removed From the Graph Neural Network, by City and Horizon, Unseen Seasons, Boston, Massachusetts (2022–2025); Columbus, Ohio (2024–2025); and Chicago, Illinois (2022–2026)

	More Error Than the Full Model, % (Seasons Clearly Worse)[a]								
	Boston (3 Seasons)			Columbus (1 Season)			Chicago (4 Seasons)		
Version	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk	1 wk	2 wk	4 wk
Graph neural network with one component removed									
No environment	0.1 (0)	−0.9 (0)	2.1 (0)	0.9 (0)	3.7 (0)	−0.3 (0)	−0.6 (0)[b]	−0.1 (0)[b]	1.7 (1)
No demographics	1.9 (1)	−0.9 (0)	0.3 (0)	−1.3 (0)	2.1 (0)	0.9 (0)	0.1 (0)	0.1 (0)	2.3 (0)
No map	10.8 (3)	3.8 (2)	5.6 (2)	15.8 (0)	6.9 (0)	1.2 (0)	7.4 (4)	6.0 (3)	3.6 (0)
Graph neural network with all 3 removed									
None of the three	9.9 (3)	4.7 (2)	5.9 (1)	16.6 (0)	7.5 (0)	0.7 (0)	8.3 (4)	5.4 (4)	2.5 (1)
History-only comparison models									
Seasonal blend	16.0 (3)	12.7 (2)	16.4 (1)	24.2 (0)	13.1 (0)	2.1 (0)	14.9 (4)	10.7 (3)	5.5 (2)
Seasonal ARIMA	27.0 (3)	18.5 (2)	11.3 (2)	36.6 (1)	16.2 (0)	3.0 (0)	11.3 (4)	6.7 (3)	2.5 (0)

Abbreviations: ARIMA, autoregressive integrated moving average; WIS, weighted interval score; wk, weeks ahead.

[a] Mean WIS of the version minus that of the full graph neural network, as a percentage of the full model's; mean WIS is the mean of the season means, as in Table 2. Positive values mean the component lowered error; negative values carry a minus sign. In parentheses, the number of seasons in which the version was clearly worse (Holm-adjusted 2-sided Diebold–Mariano $P < .05$; within the family of 4 versions for the graph neural network rows and within the comparison-model family for the 2 history-only models). The full model's relative WIS (its mean WIS divided by that of persistence) was 0.82, 0.81, and 0.72 in Boston; 0.87, 0.85, and 0.75 in Columbus; and 0.81, 0.83, and 0.83 in Chicago at 1, 2, and 4 weeks. Every version was tuned with the same budget and averaged over 10 training runs.

[b] The version without the component was clearly better in at least 1 season: in Chicago without weather, in 2 seasons at 1 week and 1 season at 2 weeks.

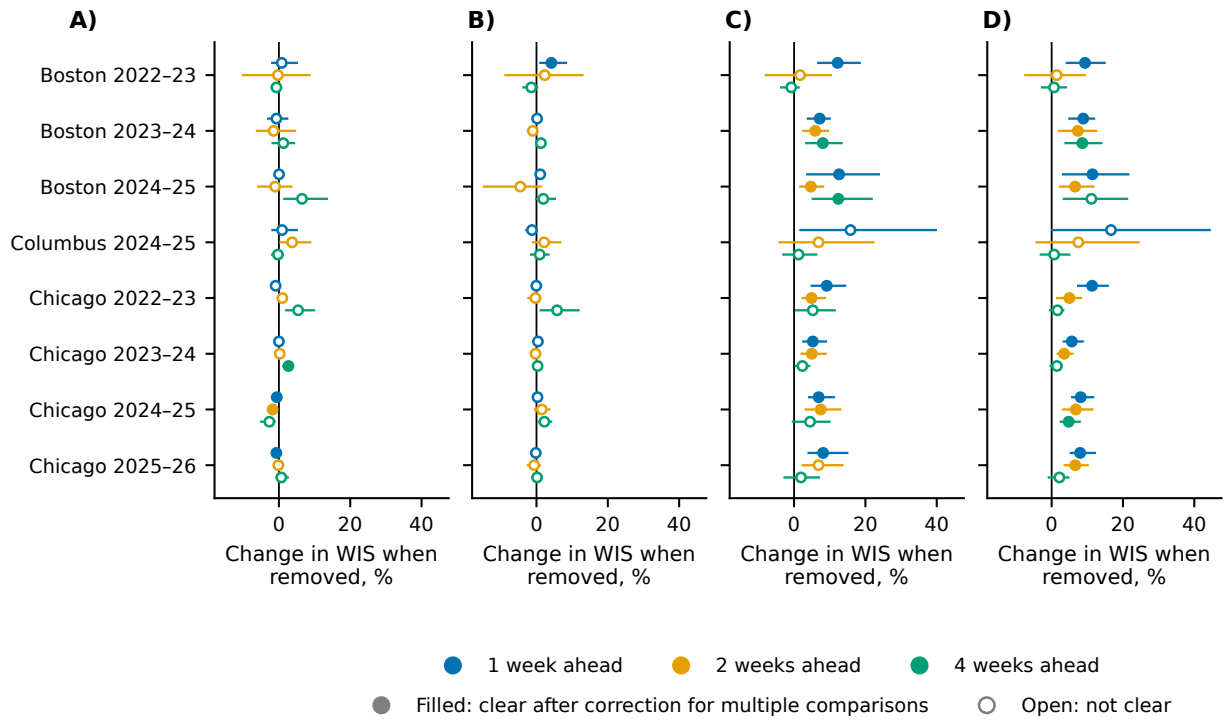


Figure 5. Change in forecast error when one component is removed from the graph neural network, by city, season, and horizon, Boston, Massachusetts (2022–2025); Columbus, Ohio (2024–2025); and Chicago, Illinois (2022–2026). A) Environment (6 weather variables). B) Demographics (8 fixed measures as inputs). C) The neighborhood map (all links between areas). D) All 3, leaving each area's own past rates. Points show the version's mean weighted interval score (WIS) minus the full model's, as a percentage of the full model's, so positive values mean the component lowered error; bars show 95% confidence intervals from a block bootstrap over weeks. Filled points are differences with a Holm-adjusted 2-sided Diebold–Mariano P value below .05, adjusted across the 4 versions within each season and horizon. Every version was tuned with the same budget as the full model.

Alt text: Four forest plots sharing rows for each city and season, each with a point and error bar for 1, 2, and 4 weeks ahead. In the environment and demographics panels, nearly all points sit near zero and are open; the few filled points are in Boston 2022–2023 (demographics, 1 week) and in Chicago, on both sides of zero for environment. In the map and all-3

panels, most points lie 4% to 12% to the right of zero and are filled at 1 week for every Boston and Chicago season; Columbus points lie to the right but are open, and 4-week points are mostly open.

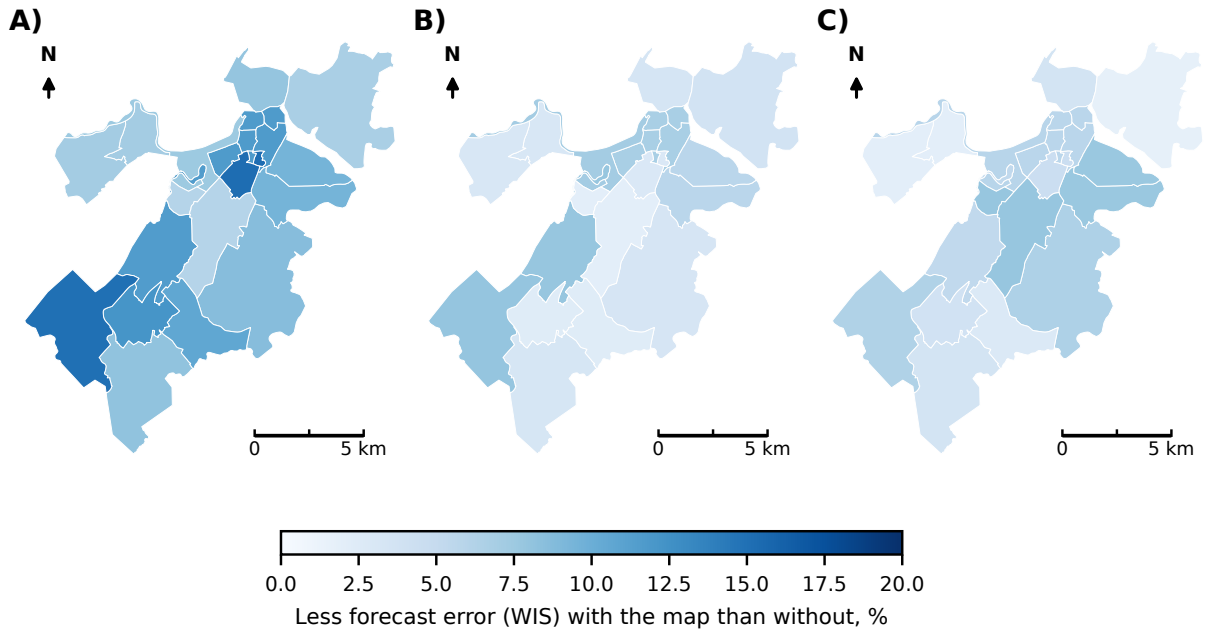


Figure 6. Reduction in forecast error from the neighborhood map, by neighborhood and horizon, Boston, Massachusetts, June 2022–May 2025. A) 1 week ahead. B) 2 weeks ahead. C) 4 weeks ahead. Shading shows the percentage by which the graph neural network's mean weighted interval score (WIS) was lower than that of the same model with the map removed, over the 3 unseen seasons. Darker blue means a larger reduction; no neighborhood had higher error with the map at any horizon.

Alt text: Three maps of Boston's neighborhoods in shades of blue. At 1 week the shading is darkest, with the largest reductions (12% to 15%) in the South End, West Roxbury, and Roslindale; at 2 and 4 weeks the shading is lighter, mostly 2% to 8%. Every neighborhood is shaded, meaning the map reduced error everywhere.

Removing weather or the demographic measures changed little (Figure 5A and B). In Boston, error changed by −0.9% to +2.1%, and 1 of 18 week-level comparisons was clear

(demographics, 2022–2023, 1 week). In Chicago, the version without demographics never differed clearly from the full model, and the version without weather was clearly better in 3 of 12 comparisons and clearly worse in 1. In Columbus, neither differed clearly.

Removing all 3 at once, so that the model saw only each area's own past rates, cost about as much as removing the map alone; relative WIS was within 0.01 of the no-map version's in every city and horizon (Figure 5D).

Did this version forecast like the comparison models that use only an area's own history? Not by the criterion we set in advance: its relative WIS was 1% to 9% below the seasonal blend's (within 5% in 4 of 9 city and horizon pairs), and it was clearly better than the seasonal blend in 12 of 24 season and horizon comparisons, never clearly worse (Appendix S1, Table S16). It also had less error than seasonal ARIMA in Boston and Columbus and was within 3% of it in Chicago, while the tuned LSTM, which reads all areas' series together, had less error than it at 1 week in Boston and Chicago. Learning from past rates adds a little; the map adds most of the rest. On the development season, weather had appeared to help slightly, which the unseen seasons did not confirm (Appendix S1, Table S17).

Transferability

Can a new city skip drawing the map by hand? The hand-drawn Boston map had 18 or 19 of its 23 borders right compared with the 2 automatically built maps, and replacing it made no clear difference in any of 36 week-level comparisons, including for a model whose only spatial information is the map (Table 4).

Table 4: Forecast Error With Hand-Drawn vs Automatically Built Neighborhood Maps, Boston, Massachusetts, 3 Unseen Seasons, June 2022–May 2025

		Mean WIS[a] by Map		
Model	Weeks Ahead	Hand Drawn	Land Boundaries[b]	Census Tracts[c]
Graph neural network	1	5.02	5.04	5.03
	2	6.28	6.24	6.39
	4	8.31	8.26	8.26
Borders only[d]	1	5.10	5.13	5.07
	2	6.38	6.39	6.33
	4	8.40	8.35	8.29

Abbreviations: ED, emergency department; WIS, weighted interval score.

[a] ED visits for influenza-like illness per 100,000 residents. Models use settings fixed before evaluation, without per-season tuning, so values differ slightly from Table 2.

[b] Official neighborhood land boundaries, matched to surrounding towns within 100 m.

[c] Neighborhoods approximated by 2020 census tracts.

[d] Uses shared borders only and learns no additional links, so the map is its only spatial information.

Can a model trained in one city forecast another? A Boston-trained model applied to Columbus with no local training had a mean WIS of 3.90, 5.08, and 6.45 at 1, 2, and 4 weeks, against 4.11, 5.18, and 6.49 for the same model trained in Columbus, with no clear difference (Table 5). In Chicago, with nearly 5 years of its own history, it was clearly worse than a local model at 1 and 2 weeks. Models moved to Boston were worse than a Boston-trained model, clearly at 4 weeks (10.07 and 10.36 vs 8.25). Every transferred model beat persistence, fine-tuning never clearly differed from training from scratch, and training on several cities at once never clearly beat a local model (Appendix S1, Table S18).

Table 5: Forecast Error When a Model Trained in One City Is Applied to Another, Boston, Massachusetts; Columbus, Ohio; and Chicago, Illinois, Unseen Season June 2024–May 2025

Target City (Source)	Weeks Ahead	Mean WIS[a]			
		Trained Locally[b]	Transferred, No Local Training	Transferred and Fine-Tuned	Persistence
Columbus (from Boston)	1	4.11	3.90	4.10	4.67
	2	5.18	5.08	5.09	5.89
	4	6.49	6.45	6.49	8.45
Chicago (from Boston)[c]	1	1.09	1.16	1.08	1.35
	2	1.12	1.24	1.12	1.42
	4	1.25	1.29	1.23	1.52
Boston (from Columbus)	1	4.88	5.28	4.78	6.36
	2	6.15	7.11	6.27	8.95
	4	8.25	10.07	8.37	14.07
Boston (from Chicago)	1	4.88	5.15	4.87	6.36
	2	6.15	6.86	6.12	8.95
	4	8.25	10.36	8.41	14.07

Abbreviations: ED, emergency department; WIS, weighted interval score.

[a] In the target city's units: ED visits for influenza-like illness per 100,000 residents in Boston and Columbus, and the health department's 1–10 activity level in Chicago. Compare columns within a row, not values across cities. The model has no neighborhood-specific learned links, so its parameters have the same size in every city, and uses settings fixed before evaluation. The source model was trained only on weeks before the target's test season.

[b] The same no-learned-links model trained on the target city alone. It is not the tuned graph neural network of Table 2, so its values differ from that table.

[c] Chicago's measure is a level rather than a rate. Each area's series is scaled by its own training-period mean and standard deviation, so the transferred model forecasts in standardized units that are converted back to levels.

Discussion

On seasons never used to design it, a graph neural network combining each neighborhood's recent trajectory, weather, demographic measures, and a map of neighboring areas forecast ED visits for ILI with 18% to 28% less error than persistence in Boston and 17% to 19% less in Chicago. Nearly all of the combination's value over a model given only past rates came from the map; removing weather or demographic measures changed error by a few percent at most. The margin over strong alternatives was modest: a tuned LSTM that sees every neighborhood's series but no map matched the graph neural network 1 week ahead in Boston, and in Columbus, with 1 unseen season, no gain was clear.

Why would the map matter when weather and demographics do not? Each area's series was scaled by its own mean, so fixed differences between neighborhoods, the kind the demographic measures describe, were absorbed before the model saw them. Neighborhoods in one city share nearly the same weather, and its seasonal part is already carried by the baseline; humidity and temperature shape when seasons occur,[24,25] but given recent rates and the time of year, little was left for them to explain. The map, by contrast, tells the model what is happening now in nearby areas, from which influenza tends to arrive.[9,10] That the LSTM also beat the version with past rates only suggests the gain comes from other areas' series more than from the particular map.

Linking areas gave little for New York City neighborhoods and for dengue in Sri Lankan districts,[21,38] whereas here the map's gain was clear at the neighborhood scale in 2 of 3 cities, including Chicago, which had 4 times as many areas and a different measure and played no part in the design. As in forecasting hubs, where gains over simple baselines have varied over time,[18,37] the advantage over the tuned LSTM varied, small in Boston but clear at 1 week in 3 of 4 Chicago seasons.

For health departments, the method needs weekly area-level rates, public boundary files, and a season definition; weather and demographic data can be left out. A department with a short history can start with a model trained elsewhere, as graph models have transferred

between countries for COVID-19,[70] though Boston and Chicago did better with their own. For alerts, a forecast's probability of crossing a threshold is more useful than its best guess, which falls short of peaks. An open-source Python package runs the forecasts and a check against persistence on a department's own data; once trained, its weekly forecasts take about a minute on a laptop (Appendix S1, Table S19). With this study's model, it reproduced the Boston evaluation within 0.4% mean WIS in all 27 model, season, and horizon comparisons;[45] by default it leaves out the demographic measures and links, which changed error little here.

This study has limitations. Components were removed one at a time, so weather and demographic measures might matter together, in cities with more varied climate, or as measures that change over time, such as vaccination coverage. Every model, like persistence, ran about h weeks behind the observed curve at h weeks ahead, so the gains come from forecasting the size of the epidemic, not its turns; earlier warning likely needs leading signals such as wastewater, available within 24 hours of sampling.[71] Columbus contributed 1 unseen season, and Chicago's measure is a capped index that flattens peaks. All models used final data, not the delayed and revised data that real-time forecasters face, which can make retrospective accuracy look better.[72] Cola-GNN was reimplemented from its description, and Boston's links across water were added by hand. Because WIS is on the scale of the rates, high-rate neighborhoods and weeks weigh most,[73] and ED visits for ILI capture people who seek emergency care, not all infections.

Next steps are a real-time test in the 2026–2027 season, now under way with weekly forecasts posted publicly (see Data Availability), ensembles of models,[74,75] and wastewater as an input.

Neighborhood-level influenza forecasts 1 to 4 weeks ahead are feasible from routine ED surveillance. Of the components we combined with each area's own history, only the map linking neighboring areas clearly lowered error; it can be built from public boundary files, and weather and demographic measures added little.

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